DIRECTORATE OF GOVERNMENT EXAMINATIONS, CHENNAI – 6
HSE SECOND YEAR EXAMINATION MARCH - 2025
PHYSICS KEY ANSWER - ENGLISH MEDIUM

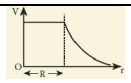
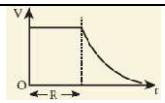
TOTAL MARKS - 70

NOTE:

1. Answer written only in **BLUE or BLOCK** should be evaluated.
2. Choose the most suitable answer in Part-I from the given alternatives and write the option code and their corresponding answer.
3. For answers in Part – II, Part – III, Part – IV like reasoning, explanation, narration, description and listing of points, students may write in their own words but without changing the concepts and without skipping any point.
4. In numerical problems if formula is not written, marks should be given for the remaining correct **steps.**
5. **In** graphical representation, physical variables for X –axis and Y-axis should be marked.

PART - I

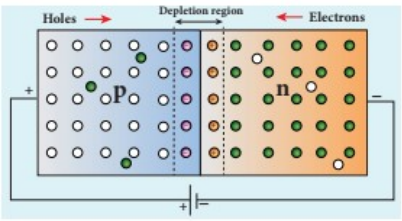
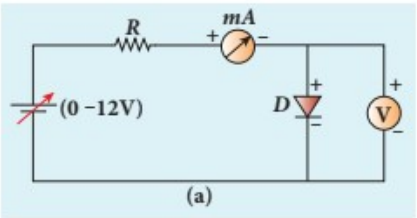
Answer **all** the Questions. **15×1=15**

Q. No	Option	Type - A	Q.No	Option	Type - B
1	d)	$\overline{A}+B+\overline{C}$	1	c)	25 cm
2	d)	0.5	2	c)	Polarisation
3	b)	2 A	3	d)	0.5
4	b)	Straight line	4	b)	2 A
5	d)	$[L^2 T^{-2}]$	5	d)	
6	a)	sports	6	d)	$\sqrt{\frac{6h\nu_0}{m}}$
7	d)	8 mC	7	a)	sports
8	b)	30^0	8	d)	8 mC
9	d)	$\frac{N_0}{\sqrt{2}}$	9	b)	30^0
10	c)	25 cm	10	d)	500 Å
11	d)	500 Å	11	d)	All the above
12	d)		12	d)	$[L^2 T^{-2}]$
13	d)	$\sqrt{\frac{6h\nu_0}{m}}$	13	b)	Straight line
14	d)	All the above	14	d)	$\frac{N_0}{\sqrt{2}}$
15	c)	Polarisation	15	d)	$\overline{A}+B+\overline{C}$

PART - II

Answer **Any SIX** Questions. Question **No 24** is compulsory **6×2=12**

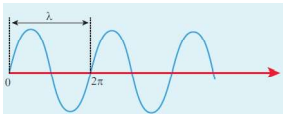
Q.No	Answer	Marks
16	$C = \frac{\varepsilon_0 A}{d}$ $= \frac{8.85 \times 10^{-12} \times 25 \times 10^{-4}}{1 \times 10^{-3}}$ $C = 22.12 \times 10^{-12} F \; or \; C = 22.12 pF$	1 ½ ½
17	The direction of the induced current always opposes the change or cause responsible for its production (or) $\varepsilon = -\frac{d\varphi_B}{dt}$ (or) $\varepsilon = -N\frac{d\varphi_B}{dt}$ ----------> 1 Mark	2
18	i) The speed of the ion is limited. ii) Electron cannot be accelerated. iii) Uncharged particles cannot be accelerated. (Any Two)	2
19	Mass defect is defined as the difference between total mass of the nucleons and actual mass of the nucleus (or) Any equivalent definition $\Delta m = (Zm_p + Nm_n) - M$	1½ ½
20	Optical path of a medium is defined as the distance *d'* light travels in vacuum in the same time it travels a distance *d* in the medium. (OR) $d' = nd$ ---------> 1 mark	2
21	a) plane wavefront b) spherical wavefront (or) plane wavefront c) cylindrical wavefront (Any one-1 mark (or) any two 1½ marks)	2

22	 (or) 	2
23	**Stopping potential** is that value of the negative (retarding) potential given to the collecting electrode *A* which is just sufficient to stop the most energetic photoelectrons emitted and make the photocurrent zero. (OR) **The minimum** negative (retarding) potential given to the anode f*or* which the photocurrent is zero.	2
24	$r = \left(\frac{\varepsilon - V}{V}\right) R$ (or) $\varepsilon = I_1(R_1 + r)$, $\varepsilon = I_2(R_2 + r)$ Substitution r = 0.5Ω	1 ½ ½

PART - III

Answer **Any SIX** Questions, Question **No.33** is compulsory **6×3=18**

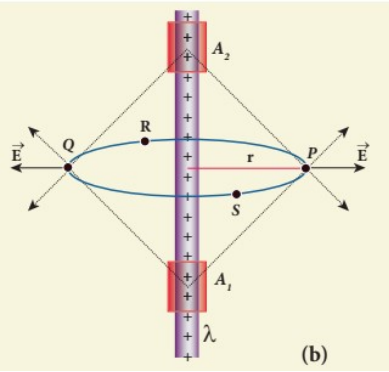
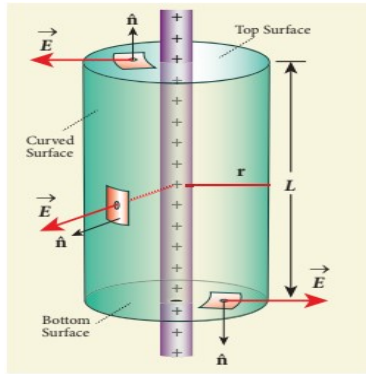
Q.No	Answer	Marks
25	It states that the algebraic sum of the currents at any junction of a circuit is zero. **(or)** ΣI =0 1 mark It states that in a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit. **(or)** Σ (IR) = Σ Ɛ 1 mark	1½ 1½
26	$\mathrm{E} = h\nu = \frac{hc}{\lambda}$ $E = \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{300 \times 10^{-9} \times 1.6 \times 10^{-19}}$ $E = 4.14\ eV$ Photoelectrons are observed.	1 1 ½ ½

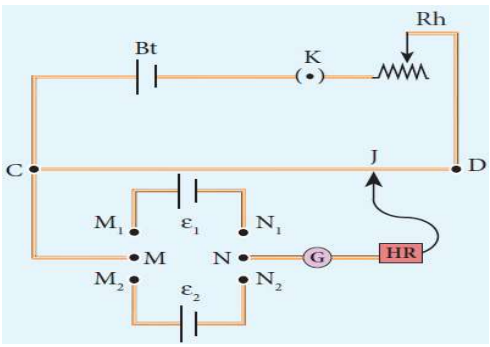
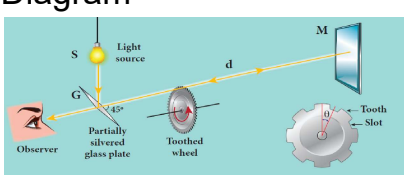
27.	When an electric charge is moving with velocity in the magnetic field , it experiences a force called magnetic Lorentz force . (or) $\vec{F}_m = q\left(\vec{v} \times \vec{B}\right)$ (or) $F_m = Bqv \sin\theta$ Any Four Features ----------> 4×½	1 2
28	**Advantages:** (i) The generation of AC is cheaper than that of DC. (ii) At higher voltages, the transmission losses are small (iii) AC can easily be converted into DC. **Any Two (2×1)** **Disadvantages:** (i) It cannot be used to charging of batteries, electroplating, electric traction etc. (ii) At high voltages, it is more dangerous to work with AC than DC. **Any One (1×1)**	2 1
29	Any **6** Properties of EM waves ----- > 6×½	3
30	Phase is the angular position of vibration when a wave is progresses, there is a relation between the phase of the vibration and the path travelled by the wave. In the path of the wave, one wavelength λ corresponds to a phase of 2π . (or)  $\delta = \frac{\lambda}{2\pi} \times \phi$ (or) $\phi = \frac{2\pi}{\lambda} \times \delta$	1 1 1
31	Any three differences between coulomb force and gravitational force -------> 3×1	3

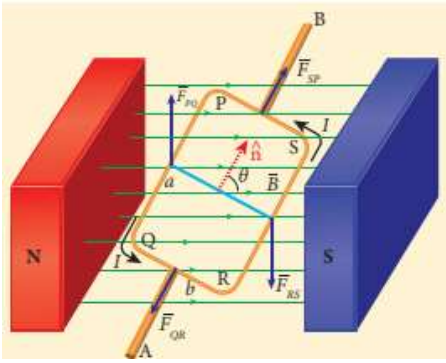
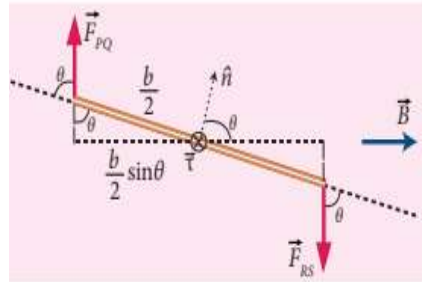
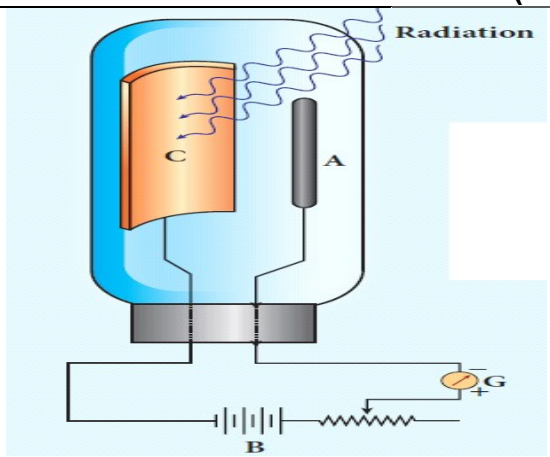
32	**Intrinsic semiconductor :** 1. Pure form of semiconductor without impurity 2. Number of electrons and holes are equal 3. Electrical conductivity low. -------> 3×½ **Extrinsic semiconductor :** 1. The semiconductor with impurity (or) The doped intrinsic semiconductor 2. Number of electrons and holes are unequal 3. Electrical conductivity high. -------> 3×½	1½ 1½
33	$\mathrm{l} = \frac{nh}{2\pi}$ $= \frac{5 \times 6.626 \times 10^{-34}}{2 \times 3.14}$ $l = 5.25 \times 10^{-34}$ kg$m^2 s^{-1}$ $\mathrm{v} = \frac{l}{mr}$ $= \frac{5.25 \times 10^{-34}}{9.1 \times 10^{-31} \times 13.25 \times 10^{-10}}$ $\mathrm{v} = 4.4 \times 10^5 \ ms^{-1}$	½ ½ ½ ½ ½ ½

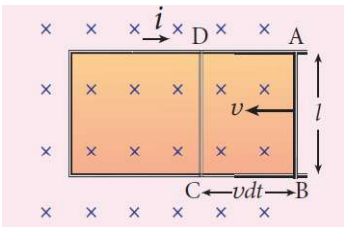
PART – IV

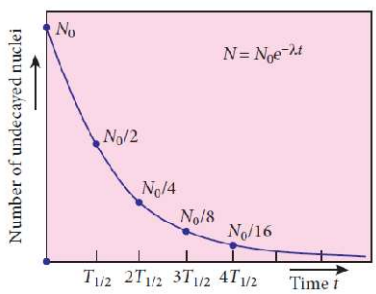
Answer **all** the questions. 5×5=25

Q.No	Answer	Marks
34 (a)	Diagram  (or)  	1
	Explanation	1
	$\Phi_E = \oint \vec{E}\cdot d\vec{A} = \int_{\text{Curved surface}} \vec{E}\cdot d\vec{A} + \int_{\text{top surface}} \vec{E}\cdot d\vec{A} + \int_{\text{bottom surface}} \vec{E}\cdot d\vec{A}$ $\Phi_E = \int_{\text{Curved surface}} E\,dA = \frac{Q_{encl}}{\epsilon_o}$ upto	1
	$Q_{encl} = \lambda L$ $E\cdot 2\pi r L = \frac{\lambda L}{\epsilon_o}$	1
	$E = \frac{1}{2\pi\epsilon_o}\frac{\lambda}{r}$ (or) $\vec{E} = \frac{1}{2\pi\epsilon_o}\frac{\lambda}{r}\hat{r}$	1
	(OR)	
34 (b)	Demorgan 1st theorem statement (or) $\overline{A+B} = \bar{A}\,.\,\bar{B}$ Demorgan2nd theorem statement (or) $\overline{A\,.\,B} = \bar{A} + \bar{B}$ 1st theorem proof with truth table 2nd theorem proof with truth table	1 1 1½ 1½

35 (a)	Diagram  Explanation $\varepsilon_1 = Irl_1$ $\varepsilon_2 = Irl_2$ $\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$ **(OR)**	1 1 1 1 1
35 (b)	Diagram  Construction and working $v = \frac{2d}{t}$ $\omega = \frac{\theta}{t}$ $\theta = \frac{total\ angle\ of\ the\ circle\ in\ radian}{number\ of\ teeth + number\ of\ cuts}$ (or) $\theta = \pi/N$ $t = \frac{\pi}{N\omega}$ $v = \frac{2dN\omega}{\pi}$	1 1 1 1 1

36 (a)	Diagram  (or)  Explanation $F_{PQ} = IaB\sin(\pi/2) = \text{I a B}$ $F_{QR} = IbB\sin(\pi/2 - \theta) = \text{I a B}\cos\theta$ $F_{RS} = IaB\sin(\pi/2) = \text{I a B}$ $F_{sp} = \text{I b B}\sin(\pi/2 + \theta) = \text{I b B}\cos\theta$ $\tau = NABI\,sin\theta \;\; (or) \;\; \vec{\tau} = (I\vec{A} \times \vec{B})$ **(OR)**	1 1 2 1
36(b)	 Construction Working Any Two uses ----------> 2×1	1 1 1 2

37(a)	(i)Diagram and explanation  $d\phi_B = \text{B x dA}$ $d\phi_B = \text{B } \ell v\, dt$ $\frac{d\phi_B}{dt} = \text{B } \ell v$ $\varepsilon = \text{B } \ell v$ ii) $\varepsilon = \frac{d\phi_B}{dt} = \frac{d(BA)}{dt} = B\frac{dA}{dt}$ $\varepsilon = \frac{0.4 \times 0.6}{1}$ $\varepsilon = 0.24$ V **(OR)**	1½ 1 ½ 1 ½ ½
37(b)	(i) Brewster's law states that the tangent of the polarising angle for a transparent medium is equal to its refractive index. **(or)** $\tan i_p = n$ 1 Mark (ii) $\tan i_p = n$ $\tan i_p = 1.65;\ i_p = \tan^{-1} 1.65;\ i_p = 58.8^0$ The inclination with the horizontal surface is, $(90^0 - 58.8^0) = 31.2^0$	2 1 1 1

38 (a)	Statement : At any instant t, the number of decay per unit time called rate of decay $\left(\frac{dN}{dt}\right)$ is proportional to the number of nuclei N at the same instant. (or) $-\frac{dN}{dt} \alpha N$ $\frac{dN}{dt} = -\lambda N$ (or) $\frac{dN}{N} = -\lambda dt$ $\int_{N_o}^{N} \frac{dN}{N} = -\int_{o}^{t} \lambda\, dt$ $ln\,[N]_{N_o}^{N} = -\lambda t\ (or)\ ln\left(\frac{N}{N_o}\right) = -\lambda t$ $N = N_o e^{-\lambda t}$ The number of atoms is decreasing exponentially over the length of time. (or)  **(OR)**	1 1 1 1 1
38(b)	When the light is allowed to pass through medium or absorbing substance then the spectrum obtained is called absorption spectrum. Each type explanation (3×1) (or) Mentioning types only 1 Mark	2 3